What Representations Preserve: Information-Theoretic and Geometric Conditions for Out-of-Distribution and Hallucination Detection

PhD Dissertation Defense

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Why confidence can be misleading

- **An unfamiliar input can still receive a confident prediction.**
  - A system trained to read chest X-rays is shown a brain MRI.
  - The system can still produce a confident diagnosis, even though the image is unlike its training examples.
- **An unsupported answer can still sound convincing.**
  - A language model produces a polished citation.
  - The cited source does not exist, but the wording alone may not reveal that.
- **The shared problem:** confidence tells us what the model chose, not whether it had the evidence needed to choose it.

What must the model preserve internally for a detector to recognize these failures?

Two lenses tell us what a detector can recover

- **A representation** is the internal description a model builds from its input.
  - A detector sees this description or a score derived from it—not the information the model already discarded.
- **Information theory asks whether the needed distinction remains.**

$$I(Z;Y) > 0 \quad \text{means that representation } Z \text{ retains some information about target } Y.$$

- **Representation geometry asks where the surviving variation lives.**
  - Is it spread across useful directions, or compressed into a narrow class-focused subspace?

Information theory tests presence; geometry tests accessibility.

Presentation Roadmap

- **Three completed studies**:
  1. **Failure—Label Blindness:** when unlabeled training removes the label distinction.
  2. **Recovery—DSC and TGT:** when supervised training suppresses domain sensitivity and guidance restores it.
  3. **Detection—Hallucination probe:** when a frozen language model retains a signal that a learned readout can expose.
- **Presentation structure**:
  - High-level overview of all three contributions.
  - Technical deep dive into the question, method, evidence, and limitations of each study.
  - Synthesis into a representation-first decision rule.
- **Unifying thesis**: detection succeeds only when the representation preserves the structure that distinguishes the failure.

Contribution 1: Label Blindness in Unlabeled OOD Detection

- **Question**: Can a representation learned without class labels preserve the distinctions needed for class-based OOD detection?
- **Conditional failure result**:
  - The input decomposes into independent feature components.
  - The surrogate task depends on one component; the ID/OOD label depends on the other.
  - A minimal sufficient representation for the surrogate contains no information about the ID/OOD label.
- **Evaluation contribution**: Adjacent OOD holds out classes from the same dataset, removing easy domain shortcuts.
- **Evidence**: unlabeled methods move toward chance on Faces, Cars, and Food while supervised MSP remains substantially stronger.

Scope: This is a theorem under explicit independence and minimal-sufficiency assumptions, not a claim that all self-supervision fails.

Contribution 2: Domain-Sensitivity Collapse and Recovery

- **Problem**: single-domain labels reward class separation but do not reward sensitivity to acquisition domain, modality, or style.
- **Diagnosis—Domain-Sensitivity Collapse (DSC)**:
  - Cross-entropy features become highly anisotropic and concentrate near a low-rank class subspace.
  - Domain-shift directions survive only in low-variance tails, weakening both distance- and logit-based scores.
- **Recovery—Teacher-Guided Training (TGT)**:
  - Distill the class-suppressed residual of a frozen multi-domain teacher during training.
  - Discard the teacher and auxiliary head at inference.
- **Evidence**: restored effective rank tracks lower FPR@95 across the evaluated scorers.

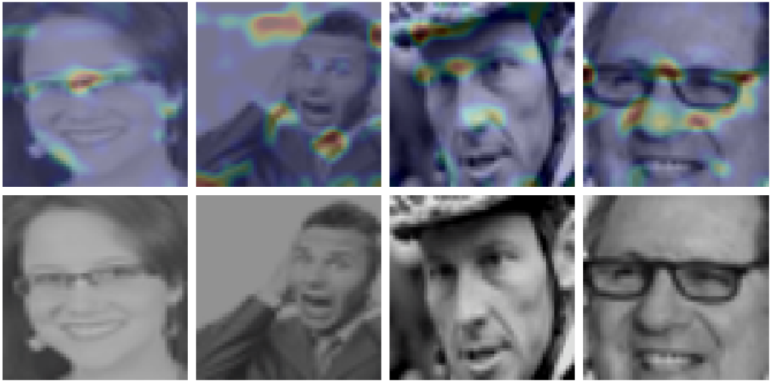
Scope: DSC provides diagnostic bounds and testable predictions; it is not an impossibility theorem.

Contribution 3: Cross-Layer Hallucination Detection

- **Question**: Does a frozen language model's residual stream retain a signal that separates truthful from incorrect short answers?
- **Method**:
  - Treat two intermediate layers from one forward pass as paired views.
  - Learn a shared compressor with asymmetric one-class contrastive supervision and log-probability reconstruction.
  - Score distance from a bank of truthful training examples.
- **Evidence**:
  - The principled probe reaches parity with the strongest engineered activation probe.
  - A theory-predicted failure of symmetric supervision appears in the ablation.
- **Access regime**: two open-weight 8B models, white-box activations, one forward pass.

Scope: The contribution is a principled and self-validating readout, not a universal or state-of-the-art claim.

Motivating Example: When Features Mislead



An unlabeled representation may emphasize easy visual regularities, such as glasses, rather than the facial-expression distinction used by the downstream label.

GradCAM is diagnostic evidence, not proof of feature independence [11].

Formal Problem Definition

- **OOD detection task**:
  - Training examples come from an in-distribution label set $\mathcal{Y}_{\mathrm{ID}}$.
  - At test time, the detector must distinguish those examples from held-out labels $\mathcal{Y}_{\mathrm{OOD}}$.
- **Two training signals**:
  - The surrogate target $Y_s = f_1(X_1)$ is available without class labels.
  - The desired ID/OOD distinction $Y = f_2(X_2)$ depends on a different feature component.
- **Label-blind setting**:

$$X = (X_1, X_2), \qquad X_1 \perp X_2,$$

  and Z is a minimal sufficient representation for Y_s.
- **Research question**: What can any detector restricted to Z recover about Y?

Information-Theoretic Analysis

- **Minimality removes information unused by the surrogate task.**

$$Y_s = f_1(X_1), \quad X_1 \perp X_2 \quad \Longrightarrow \quad I(X_2; Z) = 0.$$

- **The ID/OOD label depends only on the discarded component.**

$$Y = f_2(X_2) \quad \Longrightarrow \quad I(Y; Z) = 0.$$

- **Consequence**: every statistic computed only from Z is blind to the missing label distinction.
- **Why filtering does not help**: selecting ID classes from the population does not create dependence between the originally independent feature components.

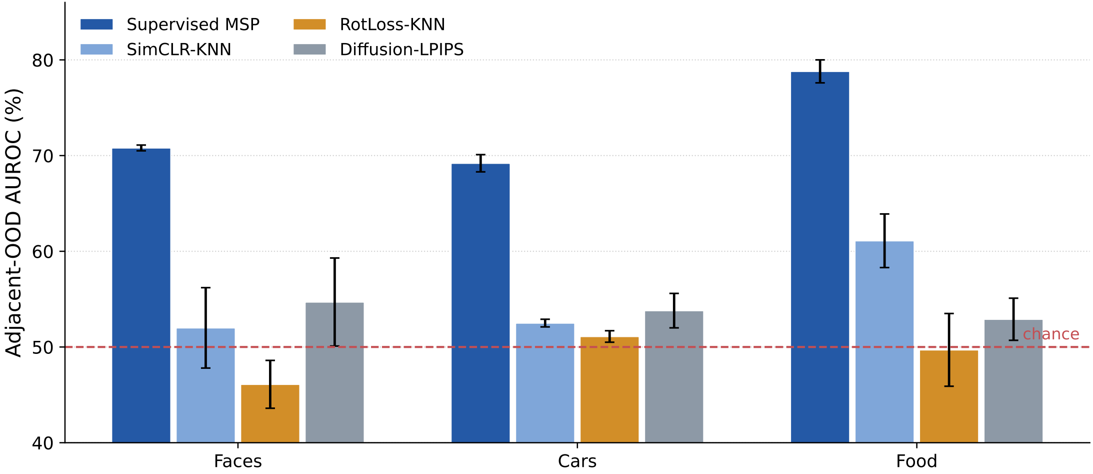
Scope: The strict result requires the stated independence and minimal-sufficiency conditions. Approximate dependence yields a warning, not the same guarantee.

Adjacent OOD Evaluation Paradigm

- **Far OOD can be solved using broad domain differences.**
  - Example: CIFAR-10 versus MNIST.
- **Near OOD reduces those differences but often changes the source dataset.**
  - Example: CIFAR-10 versus CIFAR-100.
- **Adjacent OOD uses one dataset and holds out classes.**
  - Faces, Cars, and Food share acquisition style, background statistics, and visual nuisance factors.
  - Only the semantic class distinction reliably separates ID from OOD.
- **Evaluation**: three random class splits; report AUROC and FPR@95.

High feature overlap turns label dependence into the intended stress test.

Empirical Validation Results



Unlabeled methods fall toward chance on the high-overlap task; supervised MSP preserves a clear advantage.

What the Label Blindness Result Establishes

Established

- Strict independence gives a genuine no-signal result.
- Adjacent OOD targets cases where the label distinction matters.
- A better post-hoc score cannot reconstruct information absent from Z.

Not established

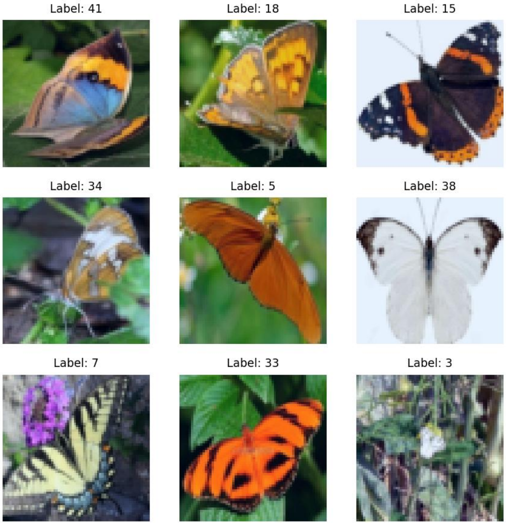
- Every self-supervised objective is label blind.
- High image overlap proves statistical independence.
- A detector with labels, external data, or another representation must fail.

The theorem identifies a boundary condition and the benchmark makes that boundary visible.

Label Blindness: Key Takeaways

- **Theoretical contribution**:
  - Formalizes when a minimal surrogate representation contains no information about the desired OOD label.
  - Connects representation sufficiency to a downstream detection limit.
- **Methodological contribution**:
  - Adjacent OOD removes domain shortcuts and tests the label-dependent distinction directly.
  - Faces, Cars, and Food expose the predicted weakness across several unlabeled method families.
- **Practical implication**:
  - Before changing the OOD score, ask whether the training objective preserved the needed distinction.
- **Status**: published at ICLR 2025.

Motivating Question: What Do We Really Learn?



A classifier learns to separate butterfly species.

Does it remain sensitive to shape, sensor, texture, and acquisition domain?

Mathematical Formalization

- **Single-domain supervised learning**: labels reward separation among C known classes inside one visual domain.
- **Feature covariance separates two sources of variation**:

$$\Sigma = \Sigma_{\text{between}} + \Sigma_{\text{within}}.$$

- **Domain-Sensitivity Collapse (DSC)**:
  - Within-class variance shrinks during cross-entropy training.
  - Most energy concentrates near the $(C-1)$-dimensional class subspace.
  - Directions that distinguish domain shift are pushed into a low-variance tail.
- **Measured signature**: low effective rank

$$r_{\text{eff}}(\Sigma) = \exp\left(-\sum_i p_i \log p_i\right), \qquad p_i = \frac{\lambda_i}{\sum_j \lambda_j}.$$

Theoretical Analysis of Collapse

- **Variance–discriminability mismatch**: the ID/OOD shift can live in directions whose ID variance is too small for ordinary distance scores to emphasize.

$$W_1(S_{\mathrm{KNN}}(Z_{\mathrm{ID}}), S_{\mathrm{KNN}}(Z_{\mathrm{OOD}})) \le L_k \varepsilon + 4 L_k \tau^2.$$

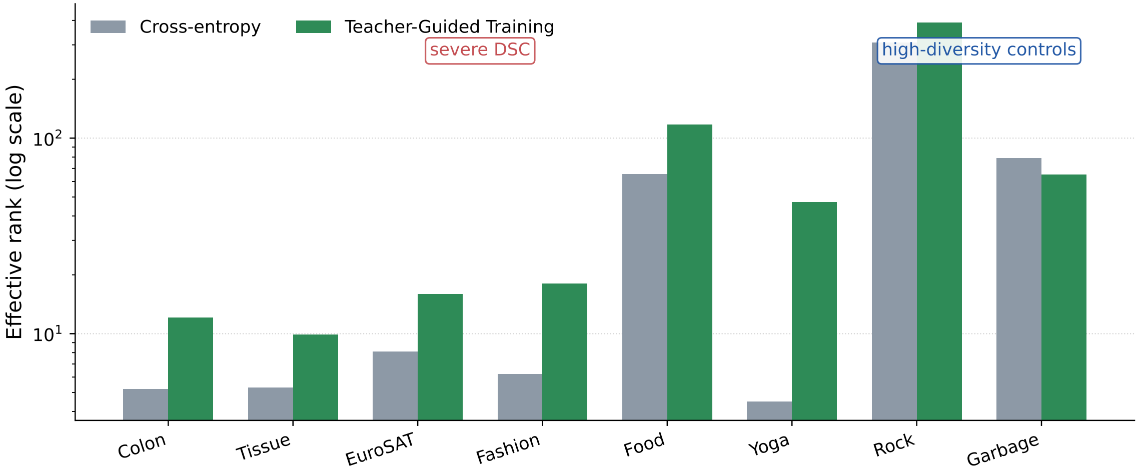
- **Head insensitivity**: if the classifier head aligns with the dominant class subspace, MSP and Energy inherit the same suppressed tail.

$$W_1(S(\ell_{\mathrm{ID}}), S(\ell_{\mathrm{OOD}})) \le \|W\|_{\mathrm{op}} \varepsilon + L_S \eta \tau.$$

- **Prediction**: low tail energy and low effective rank should coincide with weak OOD-score separation.

Scope: These are diagnostic upper bounds under measurable DSC conditions; outside that regime they may be loose or vacuous.

Empirical Demonstration of DSC



Cross-entropy features are severely anisotropic on homogeneous datasets; teacher-guided training usually restores rank.

Teacher-Guided Training Methodology

- **Teacher signal**: extract features from a frozen DINOv2 model trained across many visual domains.
- **Remove the teacher's class direction**:
  - Project out the teacher subspace most aligned with the current dataset's class labels.
  - The remaining residual emphasizes domain, style, texture, and other complementary structure.
- **Guide the student during ordinary supervised training**:

$$\mathcal{L}_{\mathrm{TGT}} = \mathcal{L}_{\mathrm{CE}} + \lambda_{\mathrm{TGT}}\, \mathbb{E}_x \left[1 - \cos\Big(\hat{h}(Z), \hat{u}_{\mathrm{dom}}(x) \Big) \right].$$

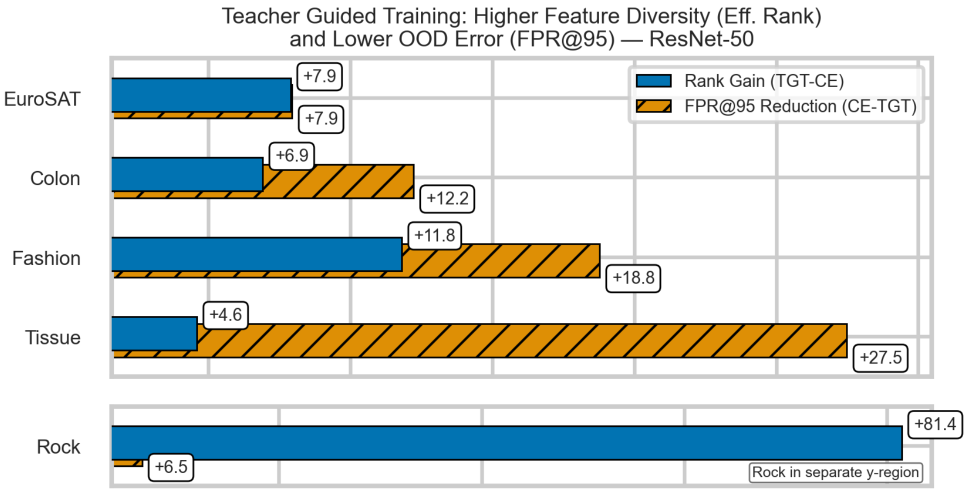
- **Inference**: discard the frozen teacher and auxiliary domain head; apply any post-hoc OOD score to the student features.

TGT changes training, not the test-time detector.

Implementation and Validation

TGT improves all eight scorers on average; the largest and most consistent reductions occur for the distance-based methods predicted by DSC.

Recovered Geometry Tracks Lower OOD Error



Across datasets, the increase in effective rank moves with the reduction in FPR@95—the proposed mechanism changes with the outcome.

DSC and TGT: Key Takeaways

- **Diagnostic contribution**:
  - DSC connects feature anisotropy and suppressed domain directions to the failure of distance- and logit-based OOD scores.
  - The theory makes testable geometric predictions rather than a universal impossibility claim.
- **Method contribution**:
  - TGT transfers a class-suppressed teacher residual during training and adds no inference overhead.
  - Mean effective rank rises from 60.2to 84.3across the eight benchmarks.
- **Limitations**:
  - Gains vary by scorer and teacher–student pairing.
  - Same-domain held-out classes remain difficult; TGT improves but does not solve that regime.
- **Status**: under review, 2026.

Information-Theoretic Hallucination Framework

- **Access regime**: a frozen language model generates an answer while the detector reads its intermediate residual-stream activations.
- **Why cross-layer views are feasible**:
  - Later residual states are deterministic functions of earlier states and intervening blocks.
  - Consequently, different layers from the same forward pass share information.
- **Trainability test**:

  $$I(Z_a; Z_b) \geq \log K - \mathcal{L}_{\mathrm{InfoNCE}}.$$

  If the views were independent, the loss could not descend below the no-information floor $\log K$.
- **Empirical burden**: detection performance must still show that the shared information contains a label-relevant slice.

Scope: Positive cross-layer dependence alone does not prove that hallucination labels are recoverable.

One-Class Contrastive Learning over Layer Pairs

- **Views**: two different layers from the same unaugmented generation, compressed by one shared network f_θ.
- **Truthful examples form the coherent reference class**:
  - Other truthful examples and the anchor's other-layer view are positives.
- **Hallucinated examples are not forced into one cluster**:
  - Only the example's own other-layer view is positive; other hallucinations remain negatives.
- **Training objective**:

$$\mathcal{L} = \mathcal{L}_{\text{SupCon-asym}} + \lambda \mathcal{L}_{\text{logprob-recon}}.$$

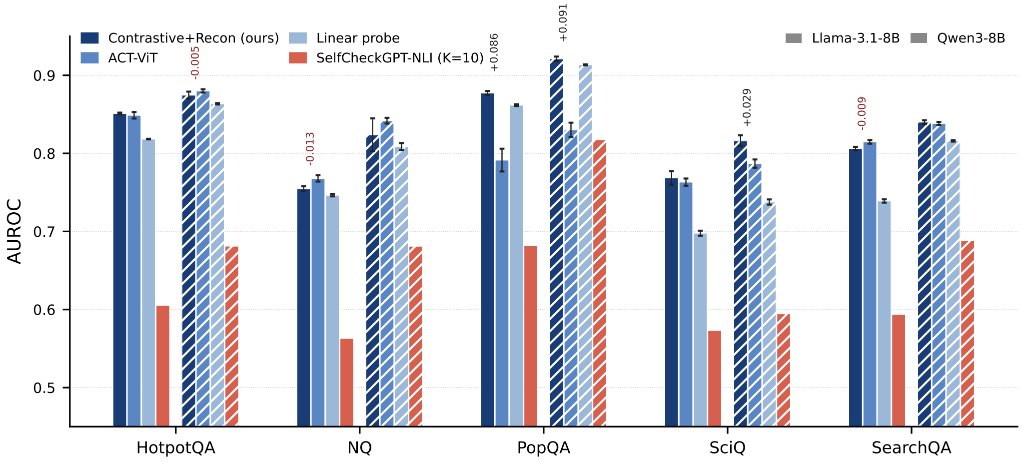
- **Prediction**: symmetric class supervision should degrade detection when hallucinations lack coherent shared structure.

System Architecture Details

- **Activation extraction**:
  - Cache full response-token activation sequences from a mid-to-late layer band.
  - No extra generation, retrieval system, or external verifier is required.
- **Shared progressive compressor**:
  - Transformer blocks reduce $4096 \to 2048 \to 1024 \to 512$ dimensions.
  - Mean pooling produces one embedding for each generation–layer view.
- **Two supervision channels**:
  - Asymmetric one-class contrastive separation.
  - Per-token log-probability reconstruction.
- **Detection**: KNN distance from the truthful training bank; the base language model remains frozen.

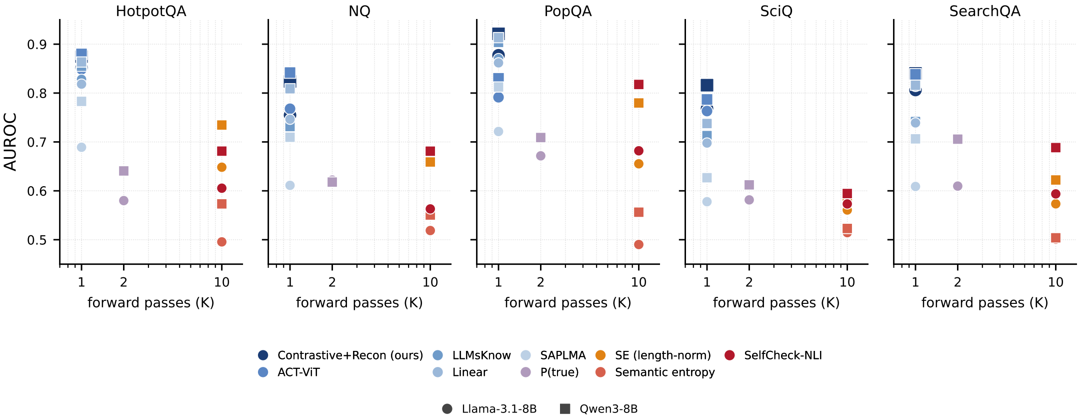
Experimental Design and Datasets

- **Models**: Llama-3.1-8B-Instruct and Qwen3-8B.
- **Short-answer QA datasets**:
  - HotpotQA, Natural Questions, PopQA, SciQ, and SearchQA.
- **Comparator families**:
  - Output-space scalars, including log-probability-based scores.
  - Single-pass activation probes, including SAPLMA and ACT-ViT.
  - $K = 10$ sampling methods, including semantic entropy and SelfCheckGPT.
- **Protocol**: train and evaluate within each model–dataset cell over five seeds; report AUROC.
- **Operational label**: an answer is a hallucination when the benchmark evaluator marks the short answer incorrect.



The principled probe reaches parity with ACT-ViT: three clear wins, three ties, one mean loss inside variance, and three clear ACT-ViT wins.

Single-Pass Activation Readers Beat Ten-Sample Methods



Activation readers use one forward pass and consistently outperform the $K = 10$ sampling family.

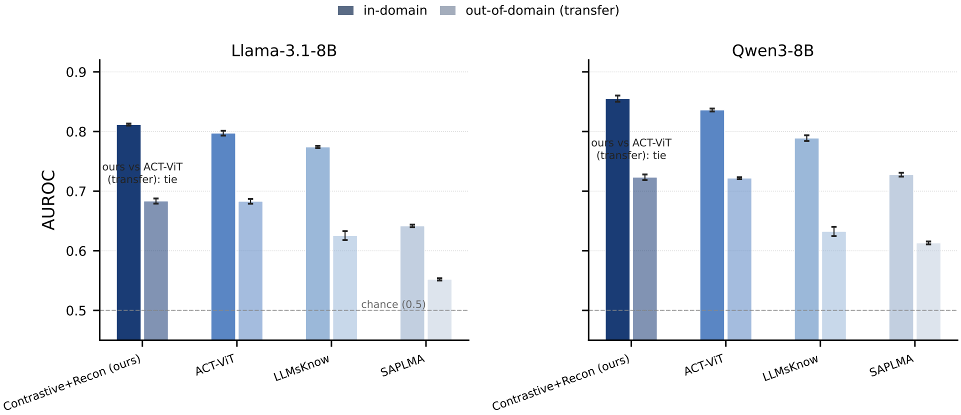
Scope: ACT-ViT shares the same one-pass advantage; this distinguishes activation access from sampling, not our probe from ACT-ViT.

Ablations Identify the Supervision That Matters

- **Reconstruction alone** adds little.
- **One-class contrast** carries nearly all of the measured gain.
- **Symmetric supervision** substantially degrades both models, as predicted.

Scope: The exact symmetric-run magnitudes remain gated pending import of the final run CSV.

The Learned Signal Transfers across QA Datasets



Cross-dataset transfer remains competitive with ACT-ViT, suggesting that the probe is not only memorizing a dataset's surface form.

Scope: Transfer is variable rather than universal; Natural Questions remains the weakest source.

Hallucination Detection: Key Takeaways

- **Feasibility contribution**:
  - Cross-layer dependence makes a contrastive readout well posed.
  - Loss descent and downstream detection show that a label-relevant slice survives compression.
- **Method contribution**:
  - Different layers of one forward pass act as paired views.
  - One-class supervision avoids inventing a coherent hallucination cluster.
- **Empirical conclusion**:
  - The method reaches parity with the strongest engineered activation probe and confirms a predicted ablation failure.
- **Limits**: white-box access, two open-weight 8B models, short-answer tasks, evaluator-defined labels, and weak Natural Questions results.

The Three Studies Address Three States of the Signal

- **Absent—Label Blindness**
  - Under strict assumptions, the learned representation contains no information about the required label distinction.
  - **Intervention**: change the objective, labels, inputs, or representation.
- **Suppressed—DSC and TGT**
  - Domain sensitivity survives only in low-variance directions that standard scores use poorly.
  - **Intervention**: repair the representation during training.
- **Retained—Cross-layer hallucination signal**
  - Intermediate activations preserve a usable distinction that output-only scores can obscure.
  - **Intervention**: learn an appropriate readout.

Failure $\longrightarrow$ recovery $\longrightarrow$ detection

The Contribution Types Have Different Claim Strengths

Study	Evidence	Defensible conclusion
Label Blindness	Conditional theorem and Adjacent-OOD stress test	A genuine failure boundary under independence and minimal sufficiency
DSC and TGT	Diagnostic bounds, geometry measurements, and intervention	Suppressed sensitivity can be diagnosed and partially restored
Hallucination	Feasibility argument, parity result, and predicted ablation	Retained internal signal supports a principled learned readout

- **Information theory** determines whether target dependence remains.
- **Representation geometry** determines how surviving variation is organized for the detector.

Limitations Define the Next Experiments

- **Label Blindness**: strict independence is a boundary condition.
  - Measure degrees of blindness and study representations that deliberately retain multiple task factors.
- **DSC and TGT**: the formal link between geometry and information loss remains incomplete; TGT inherits teacher coverage.
  - Derive linking conditions, compare complementary teachers, and target the difficult in-domain regime.
- **Hallucination detection**: the evidence is white-box and limited to short-answer QA.
  - Explain the Natural Questions failure, broaden models and tasks, and test whether the signal supports mitigation rather than detection alone.

The dissertation establishes when detection is possible and demonstrates interventions; it does not claim that these methods are optimal.

Start with the Representation before Adding Another Score

1. **Ask whether the distinction is present.**
   - If not, no post-hoc scoring rule can reconstruct it.
2. **If present, ask whether training suppressed the relevant directions.**
   - If so, repair the geometry or add complementary supervision.
3. **If accessible structure remains, design the readout around that structure.**
   - The right detector may need intermediate activations rather than output confidence alone.

A detector can use only what the representation preserves.

Thank you!

Questions?

What does the representation preserve?

Backup: OOD Metrics Used in the Dissertation

- **AUROC** $\uparrow$:
  - Probability that a randomly drawn OOD example receives a higher anomaly score than a randomly drawn ID example.
  - Chance performance is 0.5.
- **FPR@95** $\downarrow$:
  - False-positive rate on ID inputs when the threshold recovers 95% of OOD inputs.
  - This measures the deployment cost of operating at high OOD recall.
- **Usage in this dissertation**:
  - Label Blindness reports AUROC and FPR@95.
  - DSC/TGT leads with FPR@95; hallucination detection leads with AUROC.

Backup: Label Blindness Proof Chain

1. Minimal sufficiency for $Y_s = f_1(X_1)$ removes independent X_2:

$$I(X_2; Z) = 0.$$

2. Filtering the population by $Y = f_2(X_2)$ does not create dependence between the independent feature components.
3. Therefore the filtered ID representation remains independent of the target label:

$$I(Y; Z) = 0.$$

4. Any detector restricted to Z has no statistic carrying the required distinction.

Scope: Approximate label blindness uses Fano's inequality as a qualitative error warning, not the strict guarantee.

Backup: Selected Adjacent-OOD Results

Method (AUROC %)	**Faces**	**Cars**	**Food**
Supervised MSP	70.8$\pm$0.3	69.2$\pm$0.9	78.8$\pm$1.2
SimCLR-KNN	52.0$\pm$4.2	52.5$\pm$0.4	61.1$\pm$2.8
RotLoss-KNN	46.1$\pm$2.5	51.1$\pm$0.6	49.7$\pm$3.8
Diffusion-LPIPS	54.7$\pm$4.6	53.8$\pm$1.8	52.9$\pm$2.2

Backup: DSC Diagnostic Bounds

Distance failure under variance–discriminability mismatch

If the shift lies in low-variance directions and both ID/OOD tail energies are bounded,

$$W_1(S_{\mathrm{KNN}}(Z_{\mathrm{ID}}), S_{\mathrm{KNN}}(Z_{\mathrm{OOD}})) \leq L_k\varepsilon + 4L_k\tau^2.$$

MSP/Energy head insensitivity

If the classifier head is aligned with the dominant class subspace,

$$W_1(S(\ell_{\mathrm{ID}}), S(\ell_{\mathrm{OOD}})) \leq \|W\|_{\mathrm{op}}\varepsilon + L_S\eta\tau.$$

Scope: The bounds become vacuous outside the DSC regime, so they are diagnostic rather than universal guarantees.

Backup: In-Domain OOD Remains Difficult

	Out-of-domain FPR@95		In-domain FPR@95	
Scorer	CE	TGT	CE	TGT
MDS	25.55	13.94	70.12	59.07
ViM	26.43	15.65	70.12	60.68
kNN	34.98	22.71	70.28	68.39

Interpretation

TGT improves the central distance scorers, but same-domain held-out classes still overlap strongly with ID features. Recovery moves the needle without solving this regime.

Backup: TGT Objective and Inference Path

Training objective

$$\mathcal{L}_{\mathrm{TGT}} = \mathcal{L}_{\mathrm{CE}} + \lambda_{\mathrm{TGT}}\,\mathbb{E}_x\left[1 - \cos\Big(\hat{h}(Z), \hat{u}_{\mathrm{dom}}(x)\Big)\right].$$

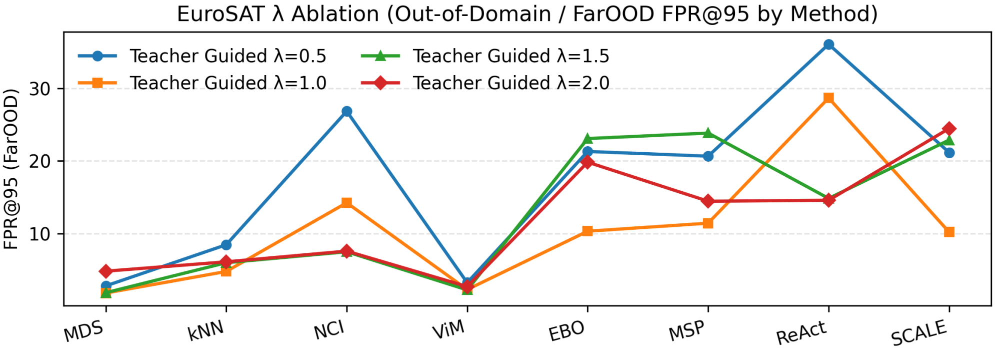
$u_{\mathrm{dom}}(x)$ is the frozen teacher feature after removing its class-discriminative projection.

Inference

$$x \longrightarrow f_s(x) \longrightarrow \text{any post-hoc OOD score.}$$

The teacher and auxiliary domain head are absent.

Backup: Teacher-Guidance Strength



Scope: The effect is scorer-dependent; distance methods are comparatively stable near the default $\lambda_{\mathrm{TGT}} = 1$.

Backup: Formal Hallucination Feasibility Argument

1. Residual computation makes later activations functions of earlier states and intervening transformations.
2. A learned compressor obeys data processing:

$$I(f_\theta(H_i); f_\theta(H_j)) \leq I(H_i; H_j).$$

3. InfoNCE lower-bounds view mutual information:

$$I(Z_i; Z_j) \geq \log K - \mathcal{L}_{\mathrm{InfoNCE}}.$$

4. Loss below the no-information floor establishes positive learned-view dependence; downstream performance separately shows that the supervised slice is label relevant.

Backup: Hallucination Evaluation Grid

- **Models**: Llama-3.1-8B-Instruct and Qwen3-8B.
- **Datasets**: HotpotQA, Natural Questions, PopQA, SciQ, SearchQA.
- **Signal families**:
  - Output-space scalar scores.
  - Single-pass activation readers.
  - Ten-sample uncertainty and consistency methods.
- **Training**: a separate probe for each model–dataset cell; five seeds.
- **Metric**: AUROC against the automatic evaluator's correct/incorrect label.

Backup: Hallucination-Study Scope

- **White-box requirement**: the detector needs intermediate activations.
- **Operational label**: incorrect short answers under an automatic evaluator are treated as hallucinations.
- **Model scope**: two open-weight 8B models; no closed API systems.
- **Task scope**: five short-answer QA datasets; no long-form, citation-grounded, or conversational generation.
- **Metrics not evaluated**: AUPRC, calibration error, and FPR@95.
- **Unresolved result**: ACT-ViT leads on Natural Questions for both models, and the cause is not yet isolated.

References I

[1] Guy Bar-Shalom, Fabrizio Frasca, Yaniv Galron, Yftah Ziser, and Haggai Maron. Beyond token probes: Hallucination detection via activation tensors with ACT-ViT. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[2] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *International conference on machine learning*, pages 1597–1607. PMLR, 2020.

[3] Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, 2024.

[4] Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. *arXiv preprint arXiv:1610.02136*, 2016.

[5] Xiaowei Huang, Daniel Kroening, Wenjie Ruan, James Sharp, Youcheng Sun, Emese Thamo, Min Wu, and Xinping Yi. A survey of safety and trustworthiness of deep neural networks: Verification, testing, adversarial attack and defence, and interpretability. *Computer Science Review*, 37:100270, 2020.

[6] Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. Supervised contrastive learning. *Advances in neural information processing systems*, 33:18661–18673, 2020.

[7] Junyi Li, Xiaoxue Cheng, Wayne Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. Halueval: A large-scale hallucination evaluation benchmark for large language models. *arXiv preprint arXiv:2305.11747*, 2023.

References II

[8] Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 3214–3252, 2021.

[9] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.

[10] Vardan Papyan, X. Y. Han, and David L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *Proc. Natl. Acad. Sci.*, 117(40):24652–24663, 2020.

[11] Ramprasaath R Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE international conference on computer vision*, pages 618–626, 2017.

[12] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. In *arXiv preprint arXiv:1807.03748*, 2018.